Sentiment-Driven Portfolio Optimization: Do Headlines Pay After Costs?

Niclas Karlsen[a,*]

[a]*Norwegian University of Life Sciences, REALTEK, Industrial Economics (INDØK)*

Abstract

This study tests whether news sentiment can improve a weekly long-only equity allocation after trading costs. It scores 661,986 company headlines with FinBERT for the members of the S&P 100 and EURO STOXX 50 and turns them into firm and sector sentiment gauges. Following CRISP-DM, all models were selected on 2012–2014 and tested once on 2015–2025 with 10 basis points per trade. Sentiment enters a Black–Litterman optimiser either as a correction of a price forecast or as the sole source of views. Sentiment signals mostly followed price moves and showed no lasting continuation. The correction did not improve forecast accuracy but raised net return by +1.92 percentage points a year. The sentiment-driven portfolio had a Sharpe ratio marginally above equal weighting, minimum variance and MSCI World but a slightly lower return, and no difference was statistically significant. The gains are small but economically relevant for a long-only investor. Headline sentiment works best as a check on a price-based allocation and is weak as a stand-alone source of alpha.

Keywords: portfolio optimization; news sentiment; FinBERT; Black–Litterman; walk-forward backtest; asset management

1. Introduction

1.1. Background

Portfolio managers, equity analysts, investment bankers and private investors follow company news closely. Language models such as FinBERT now turn thousands of headlines a week into tone scores at low cost [1, 2]. It is less clear if this helps a portfolio after costs: under semi-strong efficiency public news is priced quickly [3], while behavioural finance argues that investors under- and over-react, so tone can move prices beyond the facts reported [4, 5]. Inspired by work that feeds sentiment into allocation through reinforcement learning [6] and the Black–Litterman model [7], this study asks whether a transparent, cost-aware optimiser can let sentiment steer a realistic long-only portfolio.

1.2. Problem statement

The stakeholder is a portfolio manager of a weekly rebalanced, long-only mandate in U.S. and euro-area large caps, who must decide whether a news-sentiment layer is worth its data, computing and trading costs, and how it should enter the allocation: as a check that confirms or contradicts an existing price-based forecast, or as the driver of the allocation itself. The decision variable is the weekly weight of each stock. The key performance indicators (KPIs) are net return, Sharpe ratio, maximum drawdown and trading cost, against equal weighting, minimum variance and MSCI World. The universe is the 150 members of the S&P 100 and EURO STOXX 50 at the end of 2014 (141 with usable prices), with weekly decisions from 2015 to 2025. The results are also relevant to analysts and investors who use public headlines in their own decisions.

1.3. Research questions

Main question: *How can news sentiment be used in portfolio optimisation to improve the allocation of a weekly, long-only U.S. and euro-area equity portfolio after trading costs?*

RQ1 Does news sentiment carry information about future returns that prices have not yet absorbed, and does its effect persist or reverse?

RQ2 Does an allocation in which sentiment confirms or contradicts a price-based forecast improve return, risk and drawdown compared with the price-based allocation alone?

RQ3 Does a sentiment-driven allocation outperform equal weighting, minimum variance and the market index after costs, and how robust is the result to costs, timing and market conditions?

*Corresponding author. *E-mail address:* nikar1253@nmbu.no

1.4. Related work and knowledge gap

Table 1 summarises the closest studies, and their findings shape the design. Tone is measured against each firm's own history [8], aggregated weekly and judged over several weeks [9], and used only from the next trading session, with a delay test, because news about large firms is priced within days [10]. The reversal reported by Tetlock [5] and Hillert et al. [11] motivates testing continuation against reversal, FinBERT follows Kirtac and Germano [2], and the allocations follow Colasanto et al. [7] (sentiment as views) and Coriat and Benhamou [6] (price, combined and sentiment layers), with equal weighting as prior and benchmark [12].

Study	Data and method	Main finding
Tetlock [5]	Daily press column, word counts	Media pessimism predicts price pressure followed by reversal
Tetlock et al. [8]	Firm news, negative words	Unusual negative tone predicts lower earnings and returns
Heston and Sinha [9]	Firm news, daily vs. weekly	Weekly tone predicts returns for weeks, daily tone only for days
Ke et al. [10]	Newswires, supervised text model	News is priced within days, fastest for large firms
Hillert et al. [11]	2.2 million newspaper articles	Coverage amplifies momentum, which later reverses
Kirtac and Germano [2]	965,375 articles, FinBERT and LLMs	Transformer sentiment predicts returns better than dictionaries
Colasanto et al. [7]	BERT sentiment as Black–Litterman views	Sentiment views improved the optimised portfolio
Coriat and Benhamou [6]	Reinforcement learning with LLM sentiment	Combining price and sentiment beat either alone
DeMiguel et al. [12]	14 optimisers vs. $1/N$	Estimation error lets $1/N$ win out of sample

Table 1
Related work and its main findings.

These findings come mostly from U.S. long–short portfolios, single train–test splits or returns before costs. The gap is whether they hold inside one long-only, cost-aware optimiser for a transatlantic large-cap universe, with models selected on a separate validation period and news used only when available.

2. Theory

2.1. Market efficiency and investor psychology

Under semi-strong efficiency, a headline everyone can read should not predict returns [3]. Behavioural models disagree: investors under-react to new information, so prices drift with the news, and over-react to a series of similar news, so prices later reverse [4]. If sentiment matters, returns after an unusual shift in tone should continue or reverse.

2.2. Measuring sentiment

FinBERT is a BERT language model adapted to financial text [1]. Its financial training texts (Reuters news from 2008–2010 and a 2014 phrase bank) predate the test period, unlike those of newer large language models. For each headline it gives positive, negative and neutral probabilities, and tone is $s = p^+ - p^- \in [-1,1]$. A persistent level of tone is not news, so, following Tetlock et al. [8], a firm's weekly mean tone $\bar{s}_{i,t}$ from $n_{i,t}$ headlines is compared with the mean $\mu_{i,t}$ and standard deviation $\sigma_{i,t}$ of its own weekly tone over the previous 52 weeks,

$$a_{i,t} = \frac{n_{i,t}}{n_{i,t}+k} \cdot \frac{\bar{s}_{i,t} - \mu_{i,t}}{\sigma_{i,t}}, \tag{1}$$

where $n/(n+k)$, with $k = 5$, shrinks the score when few headlines support it ($a = 0$ without news). The same measure over all headlines of a sector gives sector sentiment. The weekly scores are smoothed into an exponentially weighted average $M_{i,t} = (1-\lambda)M_{i,t-1} + \lambda\, a_{i,t}$ with a half-life of two weeks ($\lambda = 1 - 2^{-1/2} \approx 0.29$). Divided by its typical size across firms, M becomes the *sentiment gauge* G, measured in standard deviations. A view opens only when $|G|$ passes an entry band and closes inside a narrower exit band; this hysteresis limits trading [13].

2.3. Forecasting stock returns from prices

Four families of price-based forecasts are widely used, with linear regression as the baseline: time-series models (ARIMA); factor composites of 12-month momentum [14], one-month reversal and low volatility [13]; and tree ensembles and neural networks such as LSTM, which learns from sequences of past inputs, the strongest machine-learning methods for stock returns [15]. Forecasts are judged by the rank information coefficient (IC), the weekly rank correlation across stocks between forecast and realised return. The IC measures what the optimiser uses, the ordering of stocks, and scales the expected active return [13]; squared errors such as MSE are dominated by noise in weekly returns and depend on the scale of each fore-

cast.

2.4. Portfolio optimisation with views

Mean–variance optimisation trades expected return against risk [16], but estimation error makes it fragile, and equal weighting is hard to beat [12]. The Black–Litterman model starts from neutral expected returns implied by a reference portfolio, $\pi = \delta\Sigma\mathbf{w}_{\text{ref}}$, and moves them only where the investor holds views, in proportion to their confidence [17]; without views the optimiser holds the reference portfolio, apart from trading costs. A view's size follows the Grinold–Kahn rule $\alpha_i = \text{IC} \cdot \sigma_i \cdot z_i$, so a signal with no measured skill gets no weight [13].

2.5. Conceptual model and hypotheses

Figure 1 links the constructs. A hypothesis is supported if the estimate has the expected sign and its 95% interval excludes zero, and supported in direction only if just the sign agrees.

H1 Sentiment moves prices beyond price information: after the gauge crosses its upper (lower) band, abnormal returns over the next 20 trading days are positive (negative) (H1a), and the gauge predicts next-week returns after controlling for price predictors (H1b).

H2 Correcting the price forecast with sentiment raises net return (H2a) and reduces maximum drawdown (H2b) compared with the price-only forecast, because price signals confirmed by sentiment earn more than contradicted ones (H2c).

H3 The sentiment-driven allocation has a higher Sharpe ratio after costs than equal weighting, minimum variance and the index.

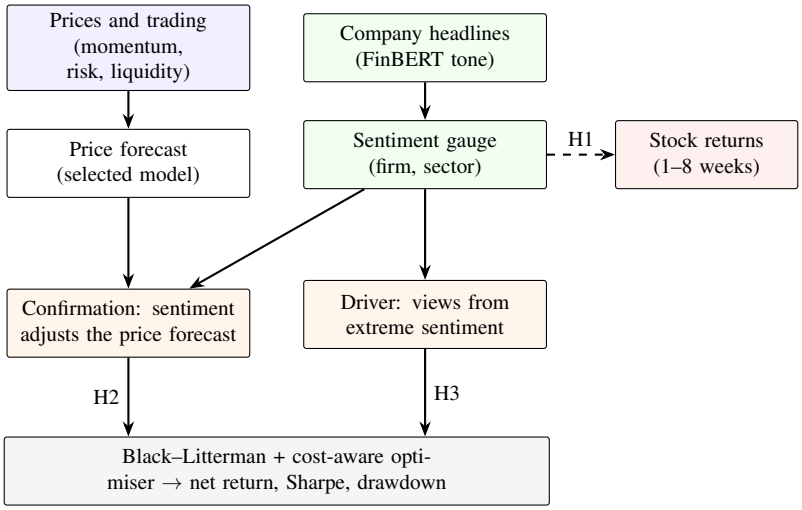


Fig. 1. Conceptual model. Price and news inputs form a price forecast and a sentiment gauge. Sentiment either confirms or contradicts the price-based forecast or drives the views itself. Labels show where each hypothesis is tested.

3. Data description

3.1. Sources and universe

All data are public. The universe is the members of the S&P 100 and the EURO STOXX 50 on 31 December 2014, the day before the test period, so membership is not chosen with hindsight. Of these 150 firms (100 and 50), nine lack a usable price history in the free data and are excluded: eight were acquired, merged or taken private between 2016 and 2025, and one has only a stale secondary listing. This leaves 141 firms, 92 in the United States and 49 in the euro area, in ten sectors (Appendix A). The exclusion uses later information; it affects 6% of the names, and the sign of the bias is unknown because takeover targets often gain when a bid is announced. Unilever is represented by its London line, as its Amsterdam line ended with the 2020 unification.

Daily prices, volume, exchange rates and the MSCI World benchmark (the URTH fund, with dividends) come from Yahoo Finance from 2005, and the risk-free rate is the three-month U.S. Treasury yield. Returns are in U.S. dollars, unhedged. Vivendi leaves the universe at its December 2024 break-up.

Headlines come from Google News searches on each company's name, one query per firm and calendar month from October 2010 to December 2025, retrieved in October 2026. Google returns at most 100 items per query, so heavily covered firms are sampled.

3.2. Preparation and quality

Figure 2 shows how 1,175,905 retrieved items become 661,986 company headlines. Duplicates are removed,

and a headline is kept only if it names the issuer. Names that are also common words or people, such as Apple or Ford, must appear together with business words. Market roundups naming three or more firms (1,789 headlines) are removed.

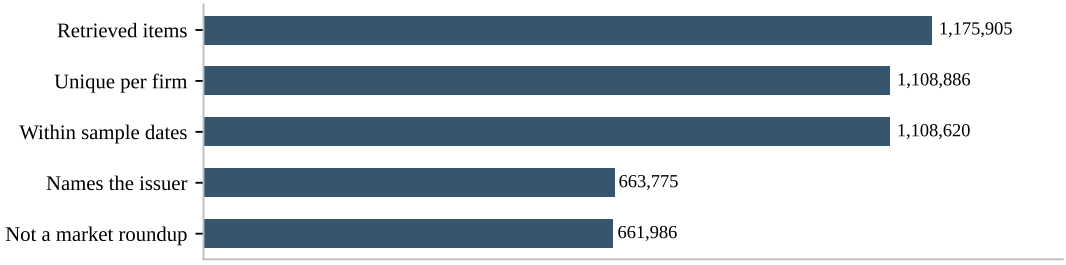


Fig. 2. Headline retention from retrieval to the weekly decision panel.

Most headlines are close to neutral, while about 16% are clearly positive and 15% clearly negative ($|s| > 0.5$; Figure 3). Euro-area headlines are more positive than U.S. headlines, and tone rose after 2020; such level differences are why tone is measured against each firm's own history.

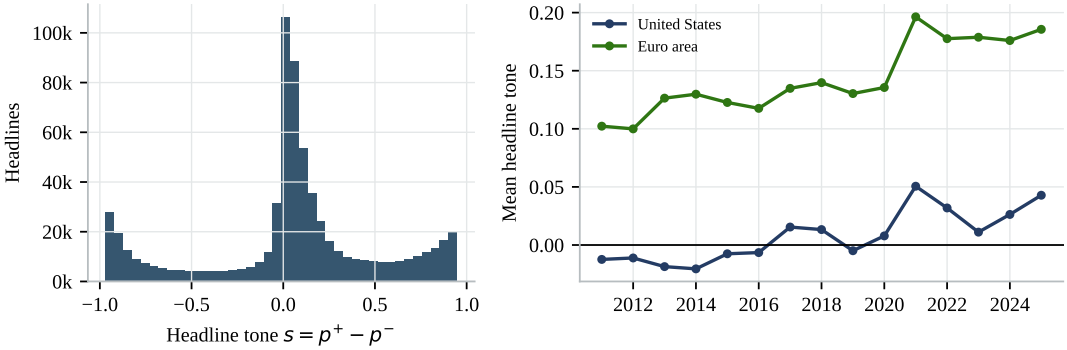


Fig. 3. FinBERT tone of the accepted headlines: distribution (left) and annual mean by region (right).

Google News reports only the publication date, so a headline counts only from the next opening of the issuer's exchange. Coverage is uneven across firms and sectors (Figure 15); the median firm has 4 headlines a week, and no news arrives in 19.7% of test stock-weeks, which then carry neutral sentiment.

4. Methodology

4.1. Research design

The study is a quantitative simulation study organised by the Cross-Industry Standard Process for Data Mining (CRISP-DM) [18] (Figure 4). Decisions are walk-forward: every Thursday at 22:00 UTC, after the U.S. close, signals are formed from data already published, and the portfolio is traded at each stock's Friday close. All strategies share one optimiser, constraints and costs, so differences come from how sentiment enters the decision. All choices were made on the validation period, 2012–2014, then fixed with a cryptographic hash of the code and settings before the test period, January 2015 to December 2025 (573 weeks), was run once (Figure 5).

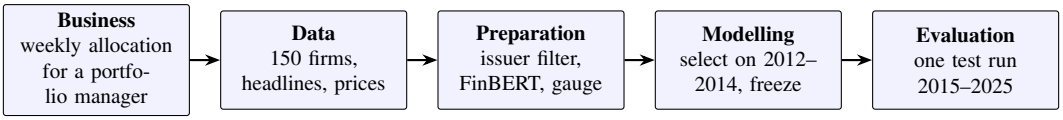


Fig. 4. Research design following the CRISP-DM phases; deployment is discussed in Section 7.

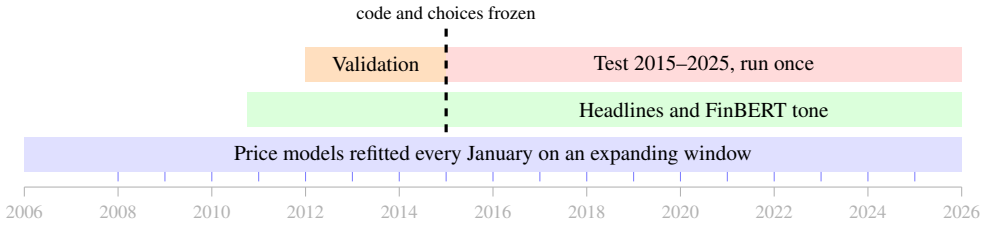


Fig. 5. Walk-forward timeline. Models are refitted every January (ticks) on data known before the first decision of the year; all choices are fixed on 2012–2014 before the test.

4.2. Price forecasting models

Six candidates represent these families: Ridge regression as the linear baseline [19], ARIMA per stock (order chosen by AIC), the factor composite, a random forest, gradient-boosted trees and an LSTM on the last eight weeks of inputs. The learning models use ten price and trading predictors (Table 2), ranked across stocks every week, and forecast returns relative to the cross-sectional mean. Fundamentals are not used because free point-in-time accounting data are unavailable for European firms. All models are pooled across stocks and refitted every January. The candidate with the highest mean weekly rank IC in the validation period is selected.

Group	Inputs (values known at the decision cutoff)
Price (7)	12-month momentum excluding the last month; 1-month and 1-week reversal; 60-day volatility; largest daily return over 20 days; beta to the universe over 52 weeks; distance to the 52-week high
Trading activity (3)	20-day dollar volume; illiquidity (mean absolute daily return per dollar traded); abnormal volume (20-day versus 120-day dollar volume)
Sentiment	abnormal firm tone and abnormal sector tone (Eq. 1), smoothed into the gauge

Table 2
Model inputs. Headline counts enter only through the shrinkage in Eq. (1).

4.3. Sentiment gauge

The gauge combines the smoothed firm and sector scores, $G = [(1-w)M^f + wM^s]/\text{s.d.}$, where the firm-only ($w = 0$) and sector-only ($w = 1$) versions are G^f and G^s. Validation chose between $w = 0$ and $w = 0.5$ by the larger absolute IC, and the sign of that IC decides whether the gauge is used for continuation or reversal. The s.d. is pooled across firms over the previous 52 weeks, which keeps the bands comparable as coverage grows. A view opens at $|G| \geq 1.5$ and closes below $|G| = 0.5$ (Figures 6 and 7).

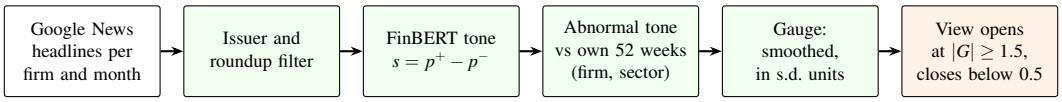


Fig. 6. From headlines to sentiment views. Each headline counts from the next session of the issuer's exchange.

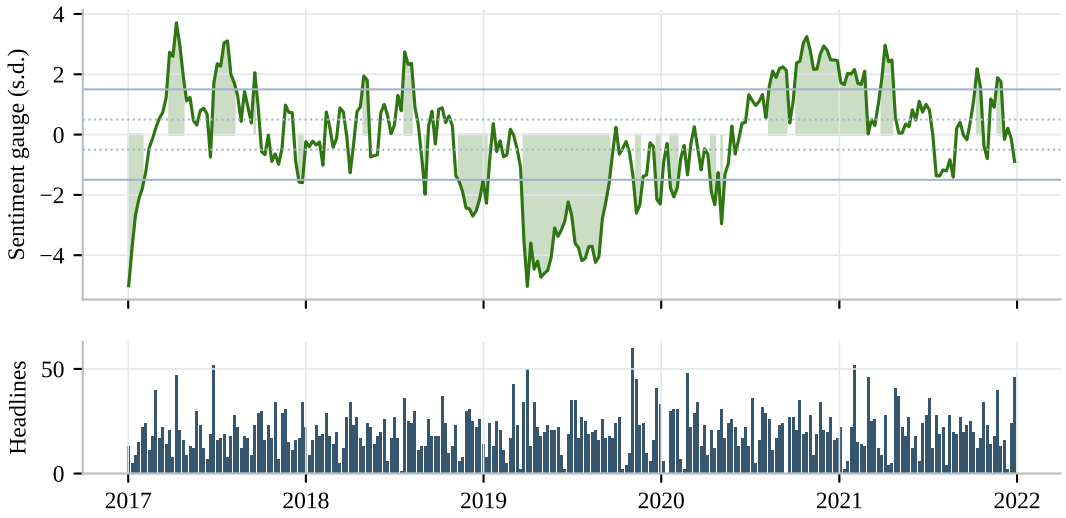


Fig. 7. Sentiment gauge and weekly headlines for Boeing, 2017–2021. Solid lines mark the entry bands and dotted lines the exit bands; shaded areas are weeks with the gauge beyond the entry band.

4.4. *Sentiment as confirmation of the price forecast*

The selected price forecast is first converted into an expected relative return $\hat{y}^P$ by a regression through the origin of realised relative returns on the forecast over the previous three years. A Ridge model then learns how sentiment corrects the forecast. To avoid look-ahead, it is trained only on *out-of-sample* errors $e = y - \hat{y}^P$, from forecasts made with earlier data, and is refitted every January:

$$\begin{aligned} \hat{y}^C_{i,t} &= \hat{y}^P_{i,t} + \beta_0 + \beta^\top \mathbf{x}_{i,t}, \\ \mathbf{x}_{i,t} &= \big(\mathbf{g}_{i,t},\, d_{i,t}\,\mathbf{g}_{i,t}\big), \\ \mathbf{g}_{i,t} &= \big(G^f, G^s, a^f, a^s\big)_{i,t}, \end{aligned} \tag{2}$$

where G^f and G^s are the firm and sector gauges, a^f and a^s the current week's abnormal firm and sector tone (Eq. 1), and $d_{i,t} = \pm 1$ the direction of the price forecast; a simpler version without the weekly tone was rejected on validation. The agreement terms $d\,\mathbf{g}$ let sentiment strengthen a price signal it confirms and weaken, or overturn, one it contradicts. The price model sets the initial decision and sentiment adjusts it. The Ridge penalty is chosen by generalised cross-validation [19].

4.5. *Sentiment as driver and portfolio optimisation*

All optimised strategies use one engine with equal weighting as the reference portfolio. A view states that stock i will beat its prior return π_i by $\alpha_i = \widehat{\mathrm{IC}}_t\,\sigma_i\,z_i$, where σ_i is the stock's weekly volatility and z_i the signal in standard deviations (the cross-sectional z-score of a forecast, or the gauge itself, capped at ± 3). The views enter the Black–Litterman posterior [17] with view uncertainty $\Omega = \tau \mathbf{D}_S$ proportional to each stock's variance, so that τ cancels:

$$\begin{aligned} \mu &= \pi + \Sigma_{\cdot S}\big(\Sigma_{SS} + \mathbf{D}_S\big)^{-1}\alpha_S, \\ \pi &= \delta \Sigma \mathbf{w}_{\mathrm{EW}}, \end{aligned} \tag{3}$$

where S is the set of stocks with views, $\Sigma_{\cdot S}$ the covariance of all stocks with them and $\mathbf{D}_S$ their variances. $\widehat{\mathrm{IC}}_t$ is the signal's mean rank IC over the previous three years of observed weeks, set to zero if negative or if fewer than 52 weeks are available. The weights then solve

$$\begin{aligned} \max_{\mathbf{w}}\ & \mu^\top \mathbf{w} - \frac{\delta}{2}\mathbf{w}^\top \Sigma \mathbf{w} - \frac{c}{H}\sum_i |w_i - \tilde{w}_i| \\ \text{s.t. } & \textstyle\sum_i w_i = 1, \quad 0 \le w_i \le 0.05. \end{aligned} \tag{4}$$

This is a convex quadratic programme in the decision variables $\mathbf{w}$ (the weekly weights). Σ is the Ledoit–Wolf covariance of the last 52 weekly returns [20], $\delta = 2.5$, $H = 4$ weeks spreads the cost of a trade over its expected holding time, and $\tilde{w}_i$ are the weights after price drift; μ and Σ are weekly. The 5% limit per issuer satisfies the UCITS rule for European funds. Three strategies differ only in their views: *price only* (views on all stocks from the price forecast), *price + sentiment* (views from Eq. 2) and the *sentiment-driven* portfolio (views only on stocks whose gauge is beyond its band). *Minimum variance* uses the same constraints without expected returns. Benchmarks are equal weighting rebalanced weekly and a buy-and-hold position in URTH.

4.6. *Costs and evaluation*

Every purchase and sale costs $c = 10$ basis points of traded value, close to the measured costs of large insti-

tutional trades [21]; 0, 5 and 20 basis points are tested. KPIs are compound annual growth (CAGR), volatility, the Sharpe and Sortino ratios (the latter counts only downside volatility; Treasury yield as risk-free rate), maximum drawdown, turnover and cost. Differences between strategies get 95% intervals from a paired bootstrap that resamples 13-week blocks to keep autocorrelation (2,000 draws) [22]; mean ICs and alphas use Newey–West (HAC) standard errors [23]. RQ1 is tested with an event study of daily returns relative to the stock's regional average around tradable gauge signals, and with Fama–MacBeth regressions of next-week returns on the gauges with and without price controls [24]. Robustness tests change one setting at a time. Two descriptive diagnostics, the hit rates in Table 6 and the return decomposition in Figure 11 with its split into buy and sell decisions, were added after the test to explain its results; they use the frozen models and change no choice.

5. Results

5.1. Model selection on the validation period

No price model forecast weekly returns reliably in 2012–2014 (Table 3, Figure 8). The factor composite had the highest mean rank IC, +0.0090 ($t = 0.39$), and was selected. The other models were close to zero or negative. The firm + sector gauge had a higher IC than the firm-only gauge (+0.0125 against +0.0039) and a positive sign, so it was used for continuation. Adding the weekly abnormal tone to the sentiment correction raised its validation IC from +0.0090 to +0.0108 (gain $t = 0.06$), so that version was kept.

Model	Mean rank IC	HAC t	Weeks IC > 0 (%)
Ridge regression	-0.0040	-0.25	49.7
Factor composite	+0.0090	0.39	54.1
ARIMA	-0.0038	-0.39	47.1
Random forest	-0.0000	-0.00	52.2
Gradient boosting	+0.0018	0.20	49.7
LSTM	-0.0031	-0.13	47.1
Sentiment gauge, firm only	+0.0039	0.60	51.6
Sentiment gauge, firm + sector	+0.0125	0.98	55.4

Table 3
Validation results, 2012–2014 (157 weeks). Rank IC with Newey–West (HAC) t-statistics; selected models in bold.

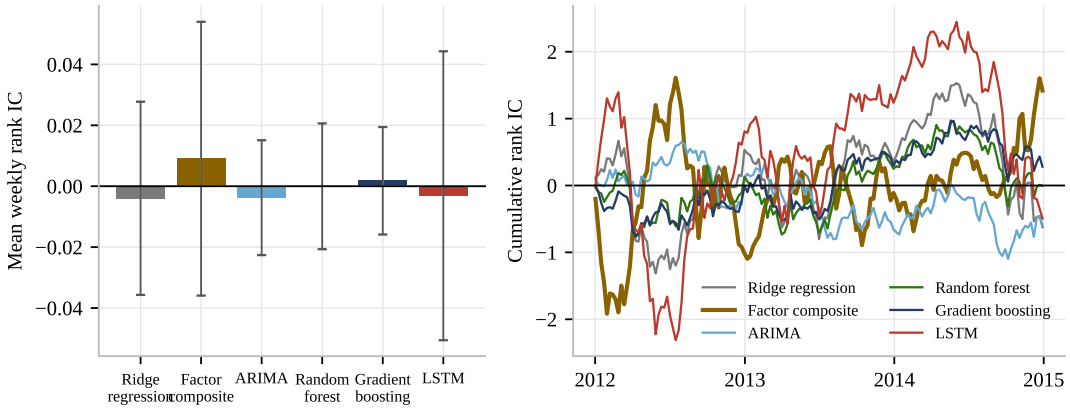


Fig. 8. Validation of the six price models: mean weekly rank IC with 95% intervals (left) and cumulative IC (right).

5.2. Sentiment and future returns (RQ1)

Figure 9 follows 1,689 positive and 1,925 negative gauge signals. Most of the abnormal return comes *before* the signal can be traded: +0.37% and -0.68% in the ten days up to day 0. After day 0, positive signals drift up by a further +0.18% over 20 trading days and then fall back (-0.36% from day 0 to day 40), while negative signals move by -0.07% over 20 days and then recover (+0.51%). The gauge mainly confirms price moves that have already happened. Over 20 days both groups move in the expected direction, but by amounts small relative to the intervals that fade by day 40, so H1a is supported in direction only.

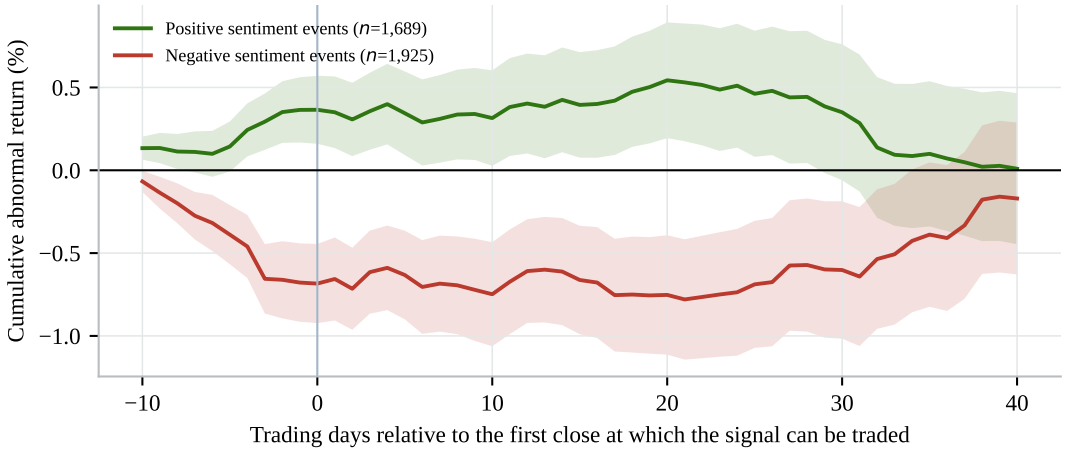


Fig. 9. Cumulative abnormal return (relative to the equal-weighted return of the stock's region) around gauge signals, 2015–2025, with 95% intervals. Day 0 is the first close at which the signal can be traded.

In Fama–MacBeth regressions (Table 4), a one-standard-deviation higher firm gauge predicts a +1.3 basis points higher relative return the next week after price controls ($t = 1.19$), +2.2 basis points in the United States ($t = 1.61$) and -0.9 in the euro area ($t = -0.45$). Sector sentiment is insignificant. H1b is supported in direction only for firm sentiment. Over horizons of one to eight weeks the gauge's rank IC is near zero, with its highest value, +0.0077 ($t = 1.11$), four weeks ahead (Figure 10).

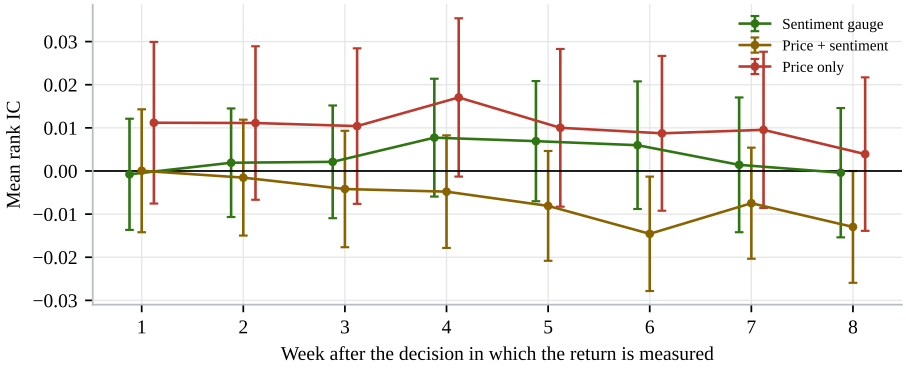


Fig. 10. Mean rank IC of each signal for returns one to eight weeks after the decision, with 95% intervals.

Regressor	Sentiment only	With price controls	U.S.	Euro area
Firm sentiment gauge	+1.1 (0.97)	+1.3 (1.19)	+2.2 (1.61)	-0.9 (-0.45)
Sector sentiment gauge	-1.9 (-0.79)	-0.8 (-0.41)	-1.1 (-0.51)	-0.9 (-0.37)
Momentum 12-1		+1.5 (0.55)	+2.5 (0.83)	+2.3 (0.72)
Reversal (1 month)		+4.9 (1.59)	+5.6 (1.47)	+3.5 (1.24)
Volatility (60 days)		-3.4 (-0.99)	-2.5 (-0.69)	-2.4 (-0.67)
Dollar volume		+3.3 (1.22)	+3.5 (1.63)	-3.9 (-1.94)
Beta (52 weeks)		+7.5 (1.49)	+4.7 (0.87)	+7.2 (1.58)
Mean R^2	0.036	0.242	0.271	0.326

Table 4
Fama–MacBeth regressions of next-week relative returns on standardised predictors, 2015–2025. Coefficients in basis points per standard deviation, Newey–West t-statistics in parentheses.

5.3. Sentiment as confirmation of the price forecast (RQ2)

In the test period the factor composite kept a small positive IC, +0.0112 ($t = 1.17$), while the corrected forecast fell to +0.0000 (Table 5). Table 6 shows the hit rates behind this result. A signal is *right* when the stock it favours beats the average stock the next week. The price signal and the sentiment gauge were each right about half of the time, whether they agreed or disagreed. When the correction moved a stock to the other side of the average (31% of cases), the original price decision would have been right more often than the new one (51.5% against 48.5%). The hit rate counts only signs, while the IC also rewards ordering large moves correctly.

Signal	Mean rank IC	HAC t	Weeks IC > 0 (%)
Factor composite (price only)	+0.0112	1.17	52.4
Price + sentiment correction	+0.0000	0.01	50.6
Sentiment gauge	-0.0008	-0.12	49.4
Correction − price only	-0.0111	-1.40	–

Table 5
Forecast quality in the test period, 2015–2025: mean weekly rank IC with Newey–West (HAC) t-statistics.

A. Did each signal pick the right side of the average?

Situation	Share of cases	Price signal right	Sentiment right
Price and sentiment agree	31%	50.0%	50.0%
Price and sentiment disagree	28%	50.6%	49.4%
Sentiment neutral	41%	50.2%	49.9%
All cases	100%	50.2%	49.6%

B. Decisions changed by the sentiment correction

	Share of cases	Original price decision right	Corrected decision right
Correction moved the stock to the other side	31%	51.5%	48.5%

Table 6
Who was right? One case is one stock in one week, 2015–2025. A signal is right when the stock it favours (penalises) does better (worse) than the average stock the next week; a coin flip is right 50% of the time. Sentiment agrees or disagrees when its gauge is at least 0.5 s.d. on the same or the opposite side of the price signal; cases with a gauge of exactly zero are left out of "Sentiment right".

The portfolio results are better than the forecast results (Table 7). The price + sentiment portfolio earned 12.20% a year with a Sharpe ratio of 0.65 and a maximum drawdown of -30.4%, against 10.28%, 0.54 and -34.4% for price only; the difference, +1.92 percentage points a year, has a 95% interval of [-1.47, 5.00] (Table 8). Figure 11 shows each signal's trailing IC over the test period and where the difference arises: as a simple average, the gain of +1.70 percentage points a year splits into +1.52 before costs and +0.17 from lower costs. The before-cost gain came equally from the price portfolio's buy decisions (stocks it overweighted; +0.76 percentage points a year, $t = 0.70$) and sell decisions (stocks it underweighted; +0.77, $t = 0.95$). When sentiment enlarged a buy or a sell, the stock moved the intended way in 52.8% and 51.9% of cases; across all adjustments the share was about one half (50.7% and 49.8%). H2a and H2b are supported in direction but not statistically, and H2c in direction only: stocks moved +0.8 basis points a week in the price signal's direction when sentiment agreed and -0.9 when it disagreed (both insignificant).

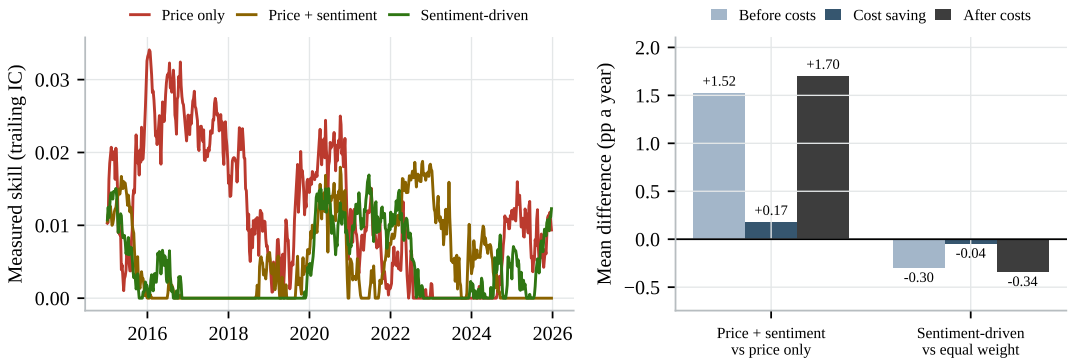


Fig. 11. Left: measured skill (trailing three-year rank IC, floored at zero) that scales each signal's views. Right: mean annual return difference before costs, from lower costs and after costs (arithmetic, percentage points).

5.4. Sentiment as the driver of the allocation (RQ3)

When sentiment drives the views alone, the portfolio held about 54 stocks with views on 20 of them in an average week; in 45% of the weeks the gauge's trailing IC was zero and the optimiser moved towards equal weighting. The portfolio earned 11.90% a year, slightly less than equal weighting (12.15%), but with a Sharpe ratio of 0.64, against 0.63 for equal weighting, 0.60 for minimum variance and 0.60 for MSCI World (Figure 12; drawdowns and annual returns in Appendix B). Its maximum drawdown, -31.6%, was smaller than that of equal weighting (-36.7%) and the index (-32.7%). All differences have bootstrap intervals that include zero (Table 8). H3 is supported in direction only.

Strategy	CAGR	Vol.	Sharpe	Sortino	Max DD	Turnover	Costs	α (t)	β
Sentiment-driven	11.90	16.77	0.64	0.89	-31.6	1.6	0.17	+1.59 (+0.67)	0.91
Price + sentiment	12.20	16.86	0.65	0.93	-30.4	3.5	0.36	+1.62 (+0.81)	0.93
Price only	10.28	16.99	0.54	0.76	-34.4	5.2	0.54	-0.11 (-0.05)	0.94
Minimum variance	9.41	13.05	0.60	0.81	-28.2	0.5	0.05	+1.15 (+0.55)	0.67
Equal weight	12.15	17.46	0.63	0.90	-36.7	1.2	0.13	+1.30 (+0.65)	0.97
MSCI World (URTH)	11.18	16.59	0.60	0.85	-32.7	–	–	–	–

Table 7
Net performance, January 2015 to December 2025 (573 weeks), 10 bp per trade. CAGR, volatility, costs and alpha in % a year; turnover is the average weekly one-way turnover in %; alpha and beta against MSCI World with Newey–West t-statistics.

Comparison (A vs B)	Return (pp a year)	Sharpe ratio	Drawdown (pp)	Significant?
Price + sentiment vs price only	+1.92 (-1.47 to +5.00)	+0.10 (-0.09 to +0.28)	+4.0 (-5.5 to +9.5)	No
Sentiment-driven vs equal weight	-0.25 (-3.23 to +2.72)	+0.01 (-0.15 to +0.14)	+5.1 (-3.9 to +9.1)	No
Sentiment-driven vs minimum variance	+2.49 (-3.09 to +8.62)	+0.03 (-0.32 to +0.34)	-3.3 (-13.9 to +2.7)	No
Sentiment-driven vs MSCI World	+0.72 (-4.01 to +5.76)	+0.03 (-0.24 to +0.29)	+1.1 (-6.2 to +10.4)	No
Price + sentiment vs equal weight	+0.06 (-2.41 to +2.56)	+0.02 (-0.12 to +0.14)	+6.3 (-4.0 to +10.7)	No
Price only vs equal weight	-1.86 (-5.20 to +1.43)	-0.09 (-0.26 to +0.08)	+2.2 (-5.0 to +6.8)	No

Table 8
Are the differences real? Each cell shows strategy A minus strategy B (for drawdown, how much smaller A's drawdown was) with its 95% confidence interval in brackets (paired block bootstrap, 13-week blocks, 2,000 draws). A difference is significant only if its interval excludes zero.

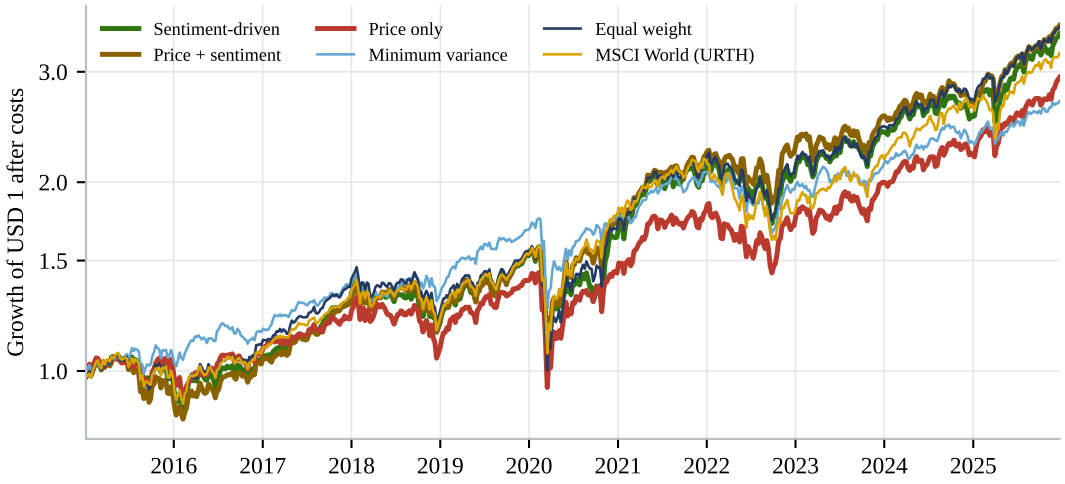


Fig. 12. Growth of USD 1 after costs, 2015–2025 (log scale).

6. Analysis

6.1. Sensitivity to costs, constraints and timing

In Table 9, the sentiment-driven portfolio's Sharpe ratio stays above equal weighting with a lower entry band of 1 s.d. (0.67), with a one- or four-week half-life, and when news reaches the portfolio one session later (0.66). The higher 2 s.d. band and the 10% position limit leave it slightly below.

The cost results are not monotonic (Figure 13). The sentiment-driven portfolio has a higher Sharpe ratio than equal weighting at every cost level, and without costs it earns 13.67% a year against 12.29%. At 20 bp all optimisers trade less, and the price-only portfolio performs best of all. A different cost rate changes the trades the optimiser makes as well as the net return, and the spread across scenarios is as large as the differences between strategies.

Scenario	Sentiment-driven	Price + sentiment	Price only	Minimum variance	Equal weight
Base case	11.90 / 0.64	12.20 / 0.65	10.28 / 0.54	9.41 / 0.60	12.15 / 0.63
Costs 20 bp per trade	12.99 / 0.70	12.72 / 0.70	13.74 / 0.75	10.41 / 0.66	12.00 / 0.62
Position limit 10%	11.73 / 0.62	12.05 / 0.64	10.67 / 0.56	8.80 / 0.56	12.15 / 0.63
Entry band 1 s.d.	12.57 / 0.67	–	–	–	–
Entry band 2 s.d.	11.48 / 0.61	–	–	–	–
Gauge half-life 1 week	12.70 / 0.68	12.30 / 0.65	–	–	–
Gauge half-life 4 weeks	12.13 / 0.65	13.02 / 0.70	–	–	–
News one session later	12.29 / 0.66	12.40 / 0.65	–	–	–
2015-2019	8.79 / 0.62	8.94 / 0.62	6.84 / 0.50	10.87 / 0.93	9.20 / 0.65
2020-2025	14.57 / 0.66	15.01 / 0.68	13.25 / 0.59	8.21 / 0.42	14.67 / 0.64

Table 9
Robustness: CAGR (%) / Sharpe ratio after costs. One setting is changed at a time; "–" means the strategy is unaffected. Costs of 0 and 5 bp are shown in Figure 13.

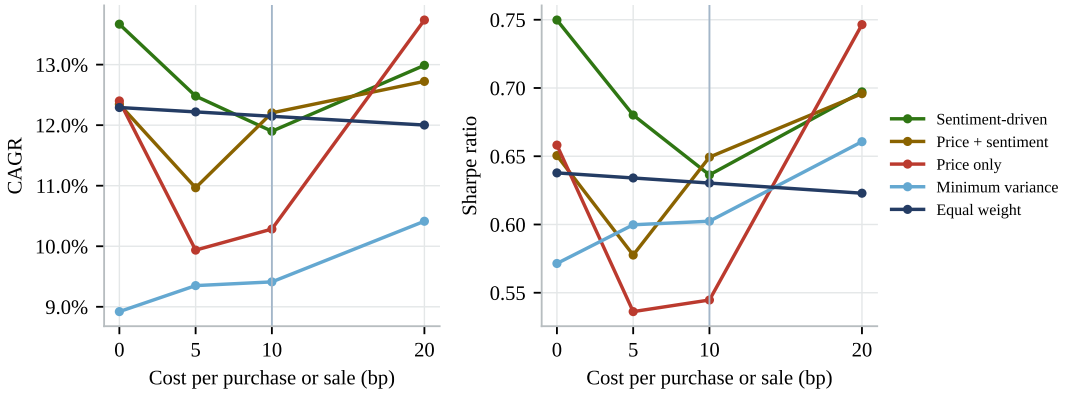


Fig. 13. Net CAGR and Sharpe ratio as functions of the cost per purchase or sale. The vertical line marks the base case.

6.2. Market conditions and sources of return

In 2015–2019 minimum variance had the highest Sharpe ratio (0.93) and the sentiment-driven portfolio trailed equal weighting slightly; in 2020–2025 the sentiment-driven and price + sentiment portfolios had the highest Sharpe ratios (0.66 and 0.68). Relative to equal weighting, the sentiment-driven portfolio returned -4.6 basis points a week in rising markets and +5.0 in falling markets (both insignificant), which suggests that its smaller drawdown comes from defensive behaviour in sell-offs.

On average the sentiment-driven portfolio held about 8 percentage points more information technology and 7 points less financials than equal weighting (Figure 14). Technology outperformed during the test period, so part of the result may come from this sector tilt rather than from picking stocks within sectors.

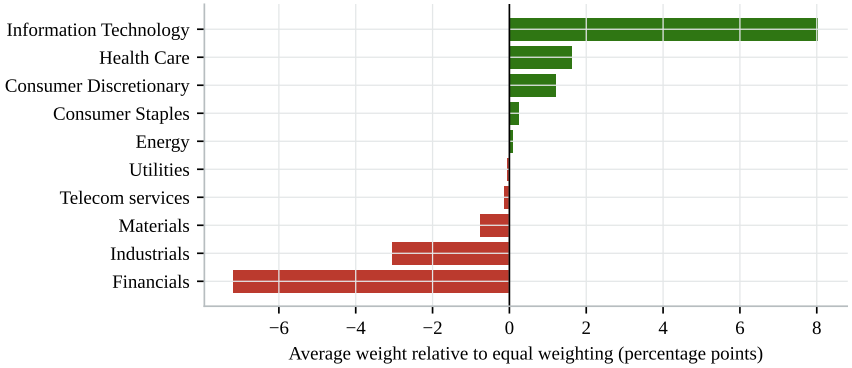


Fig. 14. Average sector weight of the sentiment-driven portfolio relative to equal weighting, 2015–2025.

7. Discussion

7.1. Interpretation and comparison with earlier studies

RQ1. Headline sentiment in large caps mostly confirms price moves that have already happened, which fits Tetlock's temporary price pressure [5] and the over-reaction to heavily covered news reported by Hillert et al. [11] better than the slow absorption of weekly tone found by Heston and Sinha [9]. The difference is consistent with Ke et al. [10], who find that news about large firms is priced within days: the firms here are heavily covered, and dates-only news reaches the portfolio a day or more after professional feeds.

RQ2. Like Coriat and Benhamou [6], the allocation that combined price and sentiment did better than the price-only allocation, with a higher return and Sharpe ratio and a smaller drawdown. Unlike in their study, the correction did not make the forecast more accurate, and the gain is not statistically significant. The gain is still economically relevant. In liquid large caps a credible edge is expected to be small and spread over many decisions [13], and an extra 1–2 percentage points a year, earned on both buy and sell decisions, matters to an investor, even if eleven years are too short to separate it from luck.

RQ3. Using sentiment as Black–Litterman views, as Colasanto et al. did [7], gave a marginally higher Sharpe ratio and a smaller drawdown than the benchmarks, with little trading. DeMiguel et al.'s warning

that $1/N$ is hard to beat [12] still holds: no difference is statistically significant over eleven years. In economic terms, a drawdown about five percentage points smaller than equal weighting at a similar return is valuable to an investor who must hold through sell-offs.

7.2. Implications for portfolio managers

The results give two guidelines. First, public headline sentiment is better used together with a price-based forecast than as a stand-alone source of alpha in liquid large caps; here it improved the allocation without improving forecast accuracy, and gains of this size justify further testing even though one sample cannot confirm them. Second, any signal should enter through a skill-weighted, cost-aware optimiser, which limits the damage when a signal stops working. Deployment is a weekly batch job of headline retrieval, FinBERT scoring and one quadratic programme, which runs in minutes on a laptop.

7.3. Limitations and future work

The universe excludes nine firms that later disappeared, a bias of unknown sign. The news archive was retrieved in 2026 and ranked by today's relevance, which may favour items still online and stories that later proved important. Many comparisons were made, eleven years give little power to detect Sharpe differences of about 0.1, the cost penalty was not tuned, and MSCI World also contains markets outside the universe. Weekly decisions cannot capture effects that last only hours. Future work should use a real-time news feed with time stamps, a broader mid-cap universe where news is absorbed more slowly, and a cost penalty and holding horizon chosen on the validation period.

8. Conclusion

This study asked how news sentiment can improve the optimised allocation of a weekly, long-only U.S. and euro-area equity portfolio after costs, with all models chosen on 2012–2014 and tested once on 2015–2025.

RQ1. Sentiment signals mostly followed price moves; the small continuation over the next month faded by day 40 (H1a in direction only). Firm sentiment had a small, insignificant positive effect after price controls (H1b in direction only).

RQ2. Correcting the factor forecast with sentiment did not raise its IC, but the portfolio earned +1.92 percentage points more a year with a smaller drawdown, a gain earned on both buy and sell decisions (H2a–H2c in direction only).

RQ3. The sentiment-driven allocation earned 11.90% a year, slightly below equal weighting, with a Sharpe ratio of 0.64, marginally above equal weighting (0.63), minimum variance (0.60) and MSCI World (0.60), and a smaller drawdown than equal weighting and the index. The differences are not significant and depend on costs and period (H3 in direction only).

Sentiment improved the allocation in direction without improving the return forecasts, and no improvement was statistically significant. Gains of one to two percentage points a year are small, but in liquid large caps they are economically relevant. For a portfolio manager, public headline sentiment is useful as a confirmation layer for a price-based allocation and weak as a source of alpha on its own.

Appendix A. Universe and news coverage

Table 10 lists the firms with their accepted headlines for October 2010–December 2025; Figure 15 shows the average number of headlines per firm and week by sector and year.

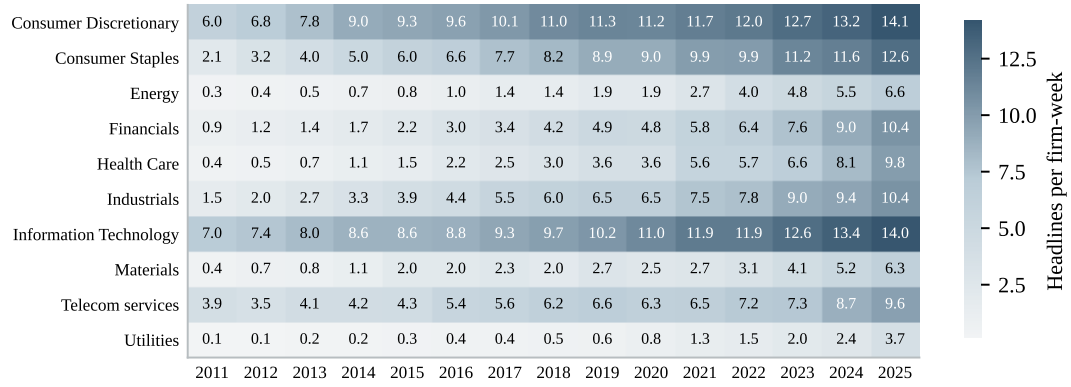


Fig. 15. Accepted headlines per firm and week, by sector and year.

Ticker	Company	Headlines	Ticker	Company	Headlines
BMW.DE	BMW	16,585	COP	ConocoPhillips	957
ITX.MC	Inditex	856	CVX	Chevron	3,815
MBG.DE	Daimler (Mercedes-Benz Group)	4,070	DVN	Devon Energy	260
MC.PA	LVMH	4,321	HAL	Halliburton	1,255
VIV.PA	Vivendi	468	NOV	National Oilwell Varco	89
VOW3.DE	Volkswagen	16,019	OXY	Occidental Petroleum	1,291
ABI.BR	Anheuser-Busch InBev	3,395	SLB	Schlumberger	1,936
BN.PA	Danone	2,057	XOM	Exxon Mobil	3,949
CA.PA	Carrefour	2,226	AIG	American International Group	3,362
OR.PA	L'Oreal	6,314	ALL	Allstate	1,912
ULVR.L	Unilever	6,547	AXP	American Express	6,054
ENI.MI	Eni	2,009	BAC	Bank of America	7,448
REP.MC	Repsol	1,394	BK	Bank of New York Mellon	1,475
TTE.PA	Total (TotalEnergies)	4,271	BRK-B	Berkshire Hathaway	3,304
ALV.DE	Allianz	2,134	C	Citigroup	6,899
BBVA.MC	BBVA	2,110	COF	Capital One	4,181
BNP.PA	BNP Paribas	2,844	GS	Goldman Sachs	9,534
CS.PA	AXA	2,032	JPM	JPMorgan Chase	8,863
DBK.DE	Deutsche Bank	4,626	MET	MetLife	3,251
G.MI	Generali	803	MS	Morgan Stanley	5,510
GLE.PA	Societe Generale	1,114	SPG	Simon Property Group	252
INGA.AS	ING Groep	390	USB	U.S. Bancorp	3,685
ISP.MI	Intesa Sanpaolo	455	WFC	Wells Fargo	7,817
MUV2.DE	Munich Re	815	ABBV	AbbVie	2,015
SAN.MC	Banco Santander	2,607	ABT	Abbott Laboratories	2,653
UCG.MI	UniCredit	935	AMGN	Amgen	1,964
BAYN.DE	Bayer	2,756	BAX	Baxter International	1,692
EL.PA	Essilor (EssilorLuxottica)	284	BIIB	Biogen	1,378
SAN.PA	Sanofi	3,112	BMY	Bristol-Myers Squibb	1,219
AIR.PA	Airbus	11,915	GILD	Gilead Sciences	2,642
DG.PA	Vinci	757	JNJ	Johnson & Johnson	6,174
DHL.DE	Deutsche Post DHL	5,565	LLY	Eli Lilly	4,391
PHIA.AS	Philips	2,641	MDT	Medtronic	3,135
SGO.PA	Saint-Gobain	775	MRK	Merck & Co.	3,991
SIE.DE	Siemens	8,343	PFE	Pfizer	6,621
SU.PA	Schneider Electric	2,080	UNH	UnitedHealth Group	1,720
ASML.AS	ASML	1,640	BA	Boeing	15,084
NOKIA.HE	Nokia	10,317	CAT	Caterpillar	2,539
SAP.DE	SAP	3,384	EMR	Emerson Electric	245
AI.PA	Air Liquide	584	FDX	FedEx	9,757
BAS.DE	BASF	2,041	GD	General Dynamics	1,542
DTE.DE	Deutsche Telekom	624	GE	General Electric	6,682
ORA.PA	Orange	70	HON	Honeywell	4,170
TEF.MC	Telefonica	1,130	LMT	Lockheed Martin	8,471
ENEL.MI	Enel	794	MMM	3M	2,017
ENGI.PA	GDF Suez (Engie)	999	NSC	Norfolk Southern	1,595
EOAN.DE	E.ON	575	RTX	United Technologies (RTX)	468
IBE.MC	Iberdrola	943	UNP	Union Pacific	2,180
RWE.DE	RWE	1,099	UPS	United Parcel Service	3,417
AMZN	Amazon.com	7,347	AAPL	Apple	15,758
CMCSA	Comcast	6,589	ACN	Accenture	5,115
DIS	Walt Disney	15,723	CSCO	Cisco Systems	7,165
F	Ford Motor	6,189	EBAY	eBay	12,534
GM	General Motors	8,799	GOOGL	Google (Alphabet)	13,809
HD	Home Depot	6,777	HPQ	Hewlett-Packard (HP Inc)	1,182
LOW	Lowe's	3,021	IBM	IBM	9,164
MCD	McDonald's	15,468	INTC	Intel	14,687
NKE	Nike	6,195	MA	Mastercard	6,149
SBUX	Starbucks	14,342	META	Facebook (Meta Platforms)	16,273
TGT	Target	6,184	MSFT	Microsoft	12,010
CL	Colgate-Palmolive	818	ORCL	Oracle	5,346
COST	Costco	10,795	QCOM	Qualcomm	6,955
CVS	CVS Health	7,747	TXN	Texas Instruments	800
KO	Coca-Cola	13,440	V	Visa	1,794
MDLZ	Mondelez International	2,436	DD	DuPont	4,819
MO	Altria	716	FCX	Freeport-McMoRan	436
PEP	PepsiCo	11,284	T	AT&T	11,195
PG	Procter & Gamble	4,939	VZ	Verizon	10,460
PM	Philip Morris International	1,825	EXC	Exelon	437
WMT	Walmart	16,087	SO	Southern Company	404
APA	Apache	87			

Table 10
Research universe: members of the S&P 100 and EURO STOXX 50 on 31 December 2014 with usable price histories, and their accepted headlines.

Appendix B. Drawdowns and annual returns

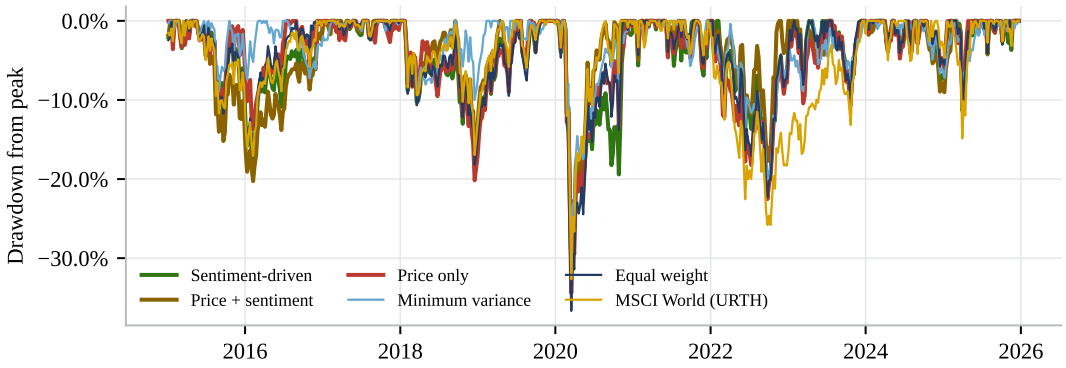


Fig. 16. Drawdown from the previous peak after costs, 2015–2025.

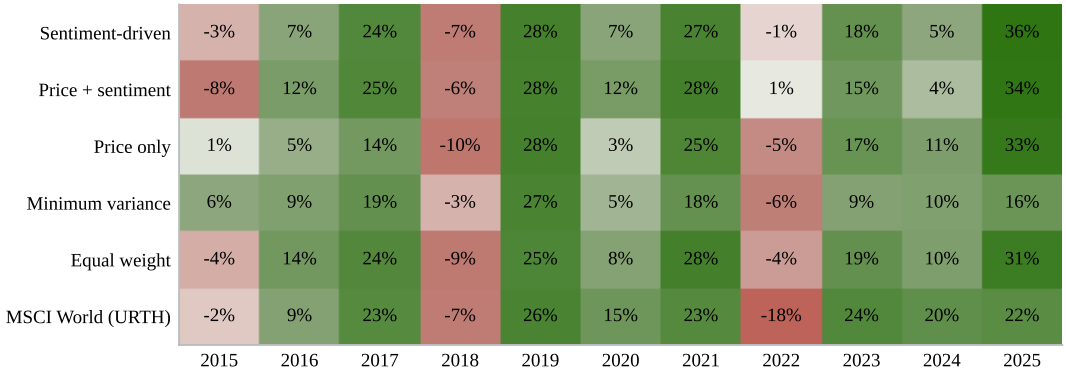


Fig. 17. Net annual returns by strategy and year.

Appendix C. Reproduction

All results, tables and figures are produced from the stored data by `project.py` in four stages: `validate-price` and `validate-sentiment` (2012–2014 only), `freeze`, which stores a SHA-256 fingerprint of the code, settings and validation choices, and `test`, which refuses to run if anything changed after freezing and runs only once. Package versions and input file hashes are recorded in `results/test_run.json`.

References

[1] D. Araci, FinBERT: Financial sentiment analysis with pre-trained language models, arXiv:1908.10063 (2019).

[2] K. Kirtac, G. Germano, Sentiment trading with large language models, Finance Research Letters 62 (2024) 105227.

[3] E. F. Fama, Efficient capital markets: A review of theory and empirical work, The Journal of Finance 25 (2) (1970) 383–417.

[4] N. Barberis, A. Shleifer, R. Vishny, A model of investor sentiment, Journal of Financial Economics 49 (3) (1998) 307–343.

[5] P. C. Tetlock, Giving content to investor sentiment: The role of media in the stock market, The Journal of Finance 62 (3) (2007) 1139–1168.

[6] B. Coriat, E. Benhamou, HARLF: Hierarchical reinforcement learning and lightweight LLM-driven sentiment integration for financial portfolio optimization, arXiv:2507.18560 (2025).

[7] F. Colasanto, L. Grilli, D. Santoro, G. Villani, BERT's sentiment score for portfolio optimization: A fine-tuned view in Black and Litterman model, Neural Computing and Applications 34 (2022) 17507–17521.

[8] P. C. Tetlock, M. Saar-Tsechansky, S. Macskassy, More than words: Quantifying language to measure firms' fundamentals, The Journal of Finance 63 (3) (2008) 1437–1467.

[9] S. L. Heston, N. R. Sinha, News vs. sentiment: Predicting stock returns from news stories, Financial Analysts Journal 73 (3) (2017) 67–83.

[10] Z. T. Ke, B. T. Kelly, D. Xiu, Predicting returns with text data, Working Paper 26186, National Bureau of Economic Research (2019).

[11] A. Hillert, H. Jacobs, S. Müller, Media makes momentum, The Review of Financial Studies 27 (12) (2014) 3467–3501.

[12] V. DeMiguel, L. Garlappi, R. Uppal, Optimal versus naive diversification: How inefficient is the $1/N$ portfolio strategy?, The Review of Financial Studies 22 (5) (2009) 1915–1953.

[13] R. C. Grinold, R. N. Kahn, Active Portfolio Management, 2nd Edition, McGraw-Hill, New York, 2000.

[14] N. Jegadeesh, S. Titman, Returns to buying winners and selling losers: Implications for stock market efficiency, The Journal of Finance 48 (1) (1993) 65–91.

[15] S. Gu, B. Kelly, D. Xiu, Empirical asset pricing via machine learning, The Review of Financial Studies 33 (5) (2020) 2223–2273.

[16] H. Markowitz, Portfolio selection, The Journal of Finance 7 (1) (1952) 77–91.

[17] F. Black, R. Litterman, Global portfolio optimization, Financial Analysts Journal 48 (5) (1992) 28–43.

[18] P. Chapman, J. Clinton, R. Kerber, T. Khabaza, T. Reinartz, C. Shearer, R. Wirth, CRISP-DM 1.0: Step-by-step data mining guide, SPSS Inc. (2000).

[19] A. E. Hoerl, R. W. Kennard, Ridge regression: Biased estimation for nonorthogonal problems, Technometrics 12 (1) (1970) 55–67.

[20] O. Ledoit, M. Wolf, A well-conditioned estimator for large-dimensional covariance matrices, Journal of Multivariate Analysis 88 (2) (2004) 365–411.

[21] A. Frazzini, R. Israel, T. J. Moskowitz, Trading costs, SSRN Working Paper 3229719 (2018).

[22] D. N. Politis, J. P. Romano, A circular block-resampling procedure for stationary data, in: R. LePage, L. Billard (Eds.), Exploring the Limits of Bootstrap, Wiley, New York, 1992, pp. 263–270.

[23] W. K. Newey, K. D. West, A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix, Econometrica 55 (3) (1987) 703–708.

[24] E. F. Fama, J. D. MacBeth, Risk, return, and equilibrium: Empirical tests, Journal of Political Economy 81 (3) (1973) 607–636.